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# Systems Engineering Roles for a New Era

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**Abstract**. Nearly thirty years ago, a classic paper was written that described systems engineering in terms of twelve roles (actually eleven plus an "other"). (Sheard, 1996) That paper is still being cited today, and the ResearchGate site reported it surpassed 20,000 reads on their site alone, including 46 citations since 2020. Yet model-based systems engineering, digital engineering, ubiquitous agile, DevSecOps, intense sociotechnical systems, and global systems of distributed architectures were nonexistent or at best newborn when that paper was written. Furthermore, an easily adopted kind of artificial intelligence appeared on the systems engineering stage around 2015 and exploded beginning in about 2020.

This paper updates the roles in the 1996 paper to provide a clearly-still-needed set of systems engineering roles consistent with today's systems engineering reality. With these roles, practitioners, managers, and those working with systems engineers have a description of what needs to be done to engineer 21st century systems and a means to negotiate who performs what tasks for which system levels. This allows a more complete set of system agreements and better operating complex systems.

**Keywords.** Systems engineering roles, digital engineering, artificial intelligence, value of roles, systems engineering challenges.

## Introduction

In the 29 years since the well-known *Twelve Systems Engineering Roles* paper (Sheard 1996) was published, the environment around systems engineering has changed drastically. Google was not even invented in 1996. Model-based systems engineering (MBSE) has grown to prominence, digital engineering has taken over from document-based engineering, and artificial intelligence has recently exploded into systems engineering practice. That systems engineering authors still cite such an old paper is evidence that the community needs a reference for critical systems engineering roles. However, we should be using a baseline of roles better adapted to this changed and changing environment. This paper presents new roles that account for industry input, software dominance, digital engineering, and sociotechnical systems, as well as the recent explosion of the use of artificial intelligence.

## *Value of Roles*

**Today's and tomorrow's problems.** INCOSE's Vision 2035 (INCOSE, 2021) calls for systems engineering solutions to meet sustainable development goals, such as ending poverty and hunger and bringing better health, education, water, and energy, to the world's people. These are major challenges that will require multiple organizations working together over a long period of time, employing many people who have top socio-technical skills, deploying many tools and processes that meet top standards, models, and compliance frameworks. Ideally, much of the knowledge embedded in systems engineering will be able to be used at the right time and place by systems engineers in rewarding careers.

**How roles have helped solve problems to date.** Some of the ways the original twelve systems engineering roles have helped solve related problems in the past are shown in Table 1. As the problems to be solved expand in scope, it is clear that updated roles are needed to contribute to solving them in the future.

Table 1. Roles Have Helped Solve Problems to Date

| **Problems to date** | **How Roles Have Helped Solve Them** |
|---|---|
| Multiple organizations work together to make bigger systems to solve bigger challenges | Organizations have agreed on who performs each of the 12 roles in which circumstances. Having a clear definition of possible roles helps organizations communicate who will do what at what level |
| Organize Systems Engineering effort across a company and between agencies | Help organizations determine what SE processes need to apply at what levels and for what purposes between different systems engineering efforts within a company and between different US Federal Agencies (Tozik, 2017) |
| Establish how standards and other models (e.g., digital engineering competency model, Scaled Agile Framework) actually impact a specific engineer) | Map these standards and models to the roles performed by organizational unit, and then assign roles to engineers (Hutchison, et al, 2021) |
| Establish SE training programs for interfacing disciplines | Show the disciplines their roles in building systems and how they fit into the SE process (Beasley, 2022) |
| Establishing career growth paths | An organization can set out typical career growth paths for engineers to consider by role. (Zavala et al., 2020, Whitcomb et al., 2015, Luna et al., 2018) |
| Make a best practice database easier to use | Tag each practice by who would use it (in which role) (Richard Turner, personal communication) |
| Defend systems engineering to standards bodies | Point out what engineers actually do by citing roles (Magarian & Seering, 2021) |

| | |
|---|---|
| Explain how to model who does what in process definition | UL Solutions (Formerly Method Park) Stages tool formally models roles within stages, giving each role specific access to process elements and tasks. (Davidz & Maier, 2007) |

# Methodology

A literature search was performed using the SCOPUS database to find papers that cited the original 12 Roles paper and the terms "revisited" or "new roles." This produced 16 papers, including all the papers analyzed in this paper, four by the same authors as the papers analyzed in this set, and six others, which were determined not central to systems engineering (for example, four pure academics without application and two that suggested tangential roles unrelated to community uses). The applicable papers were analyzed to determine what new roles were suggested, for what reasons, and how the new roles or new sets of roles compared to other suggested sets of roles.

In addition, changes in the industries that employ systems engineering were investigated using personal experience and a literature search. Comments made in the papers identified above regarding industry changes were also considered. All industry change comments were analyzed regarding their effect on roles that systems engineers should be assumed to perform. These changes were used to influence the recommended new roles in this paper.

# Eighteen Systems Engineering Roles

Table 2 below shows the 18 roles that are now considered systems engineering roles (the former roles can be found later, in Table 4). Most are consistent with systems engineering as found in SEBoK (2024) and the INCOSE Handbook (2023); the only exception, Role 11, is explained below.

**Table 2. Eighteen Systems Engineering Roles**

| | *Primary Roles* | | *Secondary Roles* |
|---|---|---|---|
| 1 CL | Coordinator/Leader | 11 AI | AI Chief |
| 2 CY | Security Engineer | 12 CH | Systems Engineering Champion |
| 3 MO | System Modeler | 13 ED | Educator |
| 4 RO | Requirements Owner | 14 IE | Implementation Engineer |
| 5 SA | System Architect | 15 IM | Information Manager |
| 6 SD | System Designer | 16 IN | Innovator/Initiator |
| 7 SI | System Interface Owner | 17 LC | Life Cycle Engineer |
| 8 ST | Stakeholder Interface Manager | 18 PO | Process Owner |
| 9 TM | Technical Manager | | |
| 10 VV | Validation and Verification Engr. | | |

The roles are systems engineering roles for the following reason: each plays a part in helping ensure a system will satisfy the stakeholders' needs, and will work well as a whole to achieve the system's purpose without unintended behaviors, in a cost-effective manner. This matches roughly what the INCOSE SE handbook characterizes as a good system (INCOSE, 2023).

This does not mean that everyone agrees these are roles for a *systems engineering division* or *group*. Companies, agencies, and programs have different names for the organizations responsible for these roles. In particular, those with software heritages and those reporting to a military hierarchy tend to have a different naming convention. Within any organization, or within any program bigger than an organization, there must be a negotiation as to who (which organization and which people) will perform what roles. Regardless of the terminology used in a given organization, however, it is possible to map the roles systems engineers play to these eighteen roles. That mapping helps with planning and scheduling of tasks, and also can enable more consistent understanding of career paths for systems engineers. (Luna et al., 2018)

Often, these roles are repeated at different levels. A government organization may be responsible for the Requirements Owner role at the systems of systems (SoS) level. A primary contractor may be responsible for the same role at the main system level. Subcontractors may be responsible for the same role at the subsystem level, and so forth for several levels down.

## *Ten Primary Systems Engineering Roles*

Primary roles are major systems engineering roles. A system is not systems engineered unless these roles are performed. The roles are frequently not performed by people called systems engineers: those performing the roles are often called software engineers, product engineers, software architects, software developers, product managers, testers, or various other names. Each primary role is shown here along with a role number and two-letter abbreviation (listed alphabetically by abbreviation). Integration of the roles is becoming critical in the new era, so primary interfaces among roles are also described.

The heritage of the roles relative to the literature is discussed in a later section (Table 4).

**1 CL Coordinator/Leader**. In this role, systems engineers are asked to coordinate diverse groups and to resolve system-level issues, to seek consensus or, if consensus cannot be achieved among the participants, to make recommendations to project management. During system design and build, whoever coordinates the implementation of system elements to specification and their delivery to systems engineering is serving in this role, from the first trade study through the last element delivery.

Leaders focus on inspiring team members and delegating work according to their preferences, strengths, and weaknesses. A good leader ensures that decisions are forward-looking, seeks a win-win solution, brings innovative approaches into the team, supports team motivation, and focuses the team on the overall project success rather than individual success. The leader also serves as a coach and mentor and shows good work ethics.

**2 CY Security Engineer.** This role is new (since 1996) and critical. Software security issues affect entire systems; system design and operation need the whole system viewpoint to prevent major damage from deliberate, planned, malevolently-intended intrusions. Cyberphysical systems now dominate the systems engineering landscape, and since “the greatest dangers [in system architecting] are at the interfaces” (Rechtin, 1991), systems engineering’s expertise in interfacing with multiple disciplines places its responsibility front and center. Per (Walden D.D., 2015, quoted in Graessler, et al, 2019): “The Security Engineer ensures that protection considerations are given a proper weight in decision making on all levels of detail and take into account security issues across the whole life cycle.” Continuing with (Graessler, 2019):

> *Therefore, the Security Engineer needs to raise awareness amongst all development team members to this issue and foster certification and accreditation. Secure engineering often deals with trade-offs; it is responsibility of this role to solve such trade-offs and create a balance between risks and opportunities.*

Mikkelsplass, et al (2023) show that AI (e.g., ChatGPT) has reduced the threshold of skill required to attack industrial control systems, so systems engineers with great cybersecurity skills are critical in this role.

**3 MO System Modeler.** System Modeler shows one of the biggest changes to the roles. The role accounts for the huge growth in the amount of computer-based modeling and simulation in systems engineering since the original paper was written. As (Græssler et al., 2019) states,

> *Modelling (sic) is more than just a Systems Engineering method. It is a core element of Systems Engineering and represents a conflation of several Systems Engineering principles. [This is] Not only a holistic perspective and consideration of system interfaces but also the top-down (decomposition) and bottom-up (integration) thinking and the continually(sic) verification & validation based on traceability.*

The System Modeler role is the curator of system models, with primarily responsibility for computer-based models, helping determine the way models will be designed, built, iterated, and integrated, including input into the modeling environment. All systems engineers will do some modeling at times, including non-computer models and computer models in their own areas of strength. This role primarily evolves from the original System Analyst role (then abbreviated SA, not the same as today’s SA, which stands for Systems Architect); see (Sheard, 1996, Hutchison, et al., 2017). The 1996 SA was never meant to be allocated to one person, but rather recognized as a role many systems engineers would play as a part of their systems engineering job.

> *“Typical analyses include system weight, power, throughput, and output predictions for hardware systems, and memory usage, interface traffic, and response times for software systems…. Modeling also helps the systems engineer and others understand how the system will be operated…. The extent of analysis varies with the type of program: the more complex and riskier programs need more analysis.”*

Written in 1996 (Sheard), this is still true, although all of these are usually modeled on computers today.

> ***Risk modeling.*** Whoever is responsible for assessing and reducing risk needs to understand the likelihood and consequences of risk. (Graessler, et al, 2019) appears to assign this to SA for early risks and Security Engineer for security risks, but is silent on program risks; one could imagine the TM takes on the risk management modeling role.

***Validation and Verification modeling***. In determining what the expected or correct answers are for every planned test will be, VV engineers will use a model of the system. Ideally, they will run the model created by the system and subsystem designers to simulate system outputs. In any case, interaction among the roles,  ensures the design process installs adequate test ports.

***Interface modeling.*** Both stakeholder interfaces and system interfaces must be modeled, by ST and SI respectively. The models allow a baseline to describe the agreements among all parties as to the details of each interface so there is no confusion. Changes then can be made as necessary, to ensure a stable set.

**4 RO Requirements Owner.** Requirements owners are responsible for basic requirements engineering and requirements management. This includes the well-known roles of requirements allocator, requirements maintainer, specification writer, development of functional architecture, and development of system and subsystem requirement from customer needs (Sheard, 1996), as well as taking on new roles, as in Table 3.

Table 3. System Requirements Owner Role Changes Related to Software Increase Since 1990s

| | New Era System Requirements Owner Roles |
|---|---|
| Agile | Write user stories and acceptance criteria; participate in sprint planning and backlog refinement |
| Agile | Participate in sprint planning and refinement of backlog |
| DevOps | Ensure requirements support automated testing and deployment needs |
| Cybersecurity | Defining cybersecurity requirements from many perspectives: vulnerabilities, threats, privacy, data breaches, and security standards |
| AI/ML | Define Data requirements for AI (Artificial Intelligence) and ML (Machine Learning) training; address AI ethics and bias requirements |
| MBSE | Ensure functional and non-functional requirements are complete; use automated requirements management and traceability tools to manage much higher complexity large-scale systems |
| (Much of this table is based on responses to prompt asked 11/6/2024 from Claude, ChatGPT 4o, and Perplexity "*What are new roles for system requirements engineers since 2000?*") | |

Often mission software is not driven by system element requirements, as was true in the 1980s; rather, mission software requirements are driven by capability requirements at the customer level. These then are allocated separately from hardware. Sometimes the interaction between system requirements and software requirements can become contentious, since hardware has a long lead time due to procurement and large complex software has a (different and possibly more variable) long lead time due to the funding, build, test, and approval process. Agile software development has done a great job in minimizing lead time, but the process of integrating that with computer-based hardware and other hardware is not always smooth.

The Requirements Owner role collaborates with the following roles:

- with subelement implementers to ensure the subelements will meet system requirements

- with SA when creating the functional architecture, to show how the functions will enable the completion of the system mission, and to use those system functions in the derivation of derived requirements.
- with MO when doing requirements modeling
- with CY when creating security requirements and ensuring they remain updated despite the fast pace of security changes
- with ST on requirements elicitation
- with SI on interface requirements
- with VV on requirements verification and validation

**5 SA Systems Architect.** An engineer in this role creates the high-level system architecture and design and selects major components. The Systems Architect generates architecture alternatives and investigates them via feasibility studies and economic analysis, compares their characteristics to the system requirements, stakeholder interests, and determines whether it is likely that subelements can be obtained that together can meet the system specification and stakeholder needs. One or more alternatives are selected and expressed in consistent views and models. A logical decomposition of the system and definition of subsystems is part of this role, to ensure feasibility of the functionality.

The Systems Architect owns the architecture concept and its realization in views and models. The Systems Architect coordinates closely with the SD, who turns the conceptual architecture into a technology-oriented design. The Systems Architect must also coordinate closely with the software systems architect to be sure that the systems design supports software architectures already in place with interfacing environmental software. If the Systems Architect isn't actively responsible for the digital engineering environment, they certainly have an integrating responsibility.

**6 SD System Designer.** The Systems Designer fleshes out architecture alternative(s) determined by the Systems Architect, turns them into specifications that together would meet the system specification and stakeholder needs, and confirms that subelements that can together meet the specifications are available or can be developed. The goal is systems-level optimization; harmonizing the different system elements and creating an overall optimum design which is superior to the performance of each element on its own. Whereas the Architect focuses on making the system work within its context, the System Designer focuses on making the system work together internally. These can be the same person on smaller projects.

Although systems engineers often do not perform detailed design, on smaller projects this might occur. Systems engineers who had played this role reported it was critical in developing their own technical and domain expertise, as well as in understanding the design approaches of classic (e.g., mechanical or electrical) engineers.

The System Designer generates various technology options satisfying the selected architecture alternative(s) and performs trade off studies, which can also be considered a MO role. With the input of the technical manager and other program personnel, the final system design is selected. The System Designer works with the domain-specific design specialists to ensure the coherence of their designs with the system and other subsystem designs. As with the SA, the System Designer must work closely with software architects to ensure that the system design supports the complex set of software architectures already in place, including environmental and test software.

**7 SI System Interface Owner.** This role is the glue that holds the system together, including not only interfaces but integration in general. This is a critical role that ensures the system vision is maintained through all the changes that occur and no issues fall through the cracks. This is not a role that could be performed by a junior person operating a systems engineering tool on a computer. The person fulfilling this role serves as the technical conscience of the program and provides the holistic perspective on any anomaly or issue. Essentially, this is the system integrator role, as defined in (Hutchison, et al., 2017).

In this role, the systems engineer is a proactive troubleshooter, preventing problems that might happen at interfaces. It is important to prevent issues between hardware and software as well as software-software and hardware-hardware issues. Today's issues are as likely to be issues between longstanding software architectures that exist around and before a given system, and system architectures that use the networks on which the software runs. In the SI role, systems engineers need wide experience, meaningful domain knowledge, and an interest in continuous learning to stay one step ahead of problems like these.

As more and more elements are selected from existing (and increasingly complex) technologies rather than built new, the role of System Interface Owner becomes more complex and more challenging.

**8 ST Stakeholder Interface Manager.** The Stakeholder Interface Manager role is responsible for ensuring the organization and the stakeholders understand clearly what the stakeholders need and what the organization is building. This includes fostering good stakeholder relationships, understanding their perspectives and viewpoints, translating their needs into well-specified requirements, and clearly communicating back to the customer which requirements (and requirements changes) have and have not been approved for inclusion in the builds. This role also may help the stakeholder understand critical technical details.

It is essential to keep requirements consistent with stakeholder needs (consistent with change approval process) and to ensure that changes are implemented thoroughly. This is a continuous task.

Systems engineers in this role may be asked to represent the point of view of the customer and to see that it is properly respected throughout the program, especially in proposal or other situations when the organization may be prohibited from talking to the customer.

Agile software engineering depends to a great extent on close relationships with stakeholders, particularly customers and users, through user stories and discussions of pain points and feature value. Unlike marketing roles, the systems engineering ST role focuses on technical aspects.

Stakeholder interface manager works with RO to ensure requirements are properly elicited and then fully managed and tracked over the life cycle.

**9 TM Technical Manager.** Technical management is one part of program management. A Technical Manager is asked to be responsible for technology-based decisions regarding the *system of interest* while a non-technical project manager oversees cost, schedule, and resources (Græssler et al, 2019). Both broad and deep knowledge about the included domains is needed. The Technical Manager needs to ensure information continues to flow by strongly involving other systems engineering roles.

With increased globalization, development team size is increasing, team members come from different countries, and work may occur at different locations. A Technical Manager must coordinate efforts, align them with the overall goal, and match team members' knowledge and skills. By representing the project hub, this role should also proactively identify problems and prevent them. Such problems may be technical, organizational, or personal.

In commercial companies, a type of systems engineer called the Product Manager or Product Engineer often appears. This TM role is similar to an aerospace/defense *Systems Engineering Manager* role, with authority over a much smaller group of engineers, maybe only one. On a small project, the Product Manager or Product Engineer may wear more of a marketing, cost, and schedule hat than technical managers on large programs wear.

The role in particular coordinates with ST, stakeholder interface management, for external negotiations and supplier management, with software leadership, and with CL, for coordination across the organization.

**10 VV Validation and Verification Engineer.** This role is responsible for measuring system performance (Verification), and assuring stakeholder needs fulfilment (Validation). VV engineers plan and implement, or in different implementations oversee, the system verification program, including testing, demonstration, and simulation, to ensure the system as designed and built will meet its requirements. In some organizations, systems engineers also create the detailed system test plans and test procedures. VV engineers also are required to respond to anomalies with the best possible understanding of the system design, and to know which experts to call when needed.

The VV Engineer interacts with the RO closely to see that the fulfilment of requirements is measured appropriately and ensures stakeholder satisfaction, and with software engineers to coordinate software testing during the V&V activities. The VV role also works continuously on the digital engineering environment either as part of the system under consideration or as part of an interfacing system.

## *Eight Secondary Systems Engineering Roles*

Secondary roles are those that support systems engineering and can be done independent of actually engineering a system. While not always called systems engineering roles, they fundamentally qualify in some way as ensuring some system works well together; in this case, the system may be the enterprise or the process of engineering systems. These roles are listed here.

**11 AI AI Chief.** Cummings (2024) made a strong case that a Chief AI Risk/Safety/Teaming officer should be a required role in any organization working on programs today, or at least in those organizations making products or performing services for the government. It is reasonable to make this at least a secondary systems engineering role, as AI (and machine learning) can and most likely will affect all areas of the system, much as modeling affects all areas of the system. AI is moving too fast for anyone not working on it full time to comprehend properly, and at this time most systems and software engineers are nowhere near being AI experts. Eventually it will become as much a commodity as computers are, but during this transitional period its introduction, use, and publicity should be intelligently engineered. (There may be a time in the next five years when that AI Chief can be folded in under IM or TM, or it may be considered under something other than systems engineering.)

**12 CH Systems Engineering Champion.** The systems engineering champion promotes the value of systems engineering goals, tasks, and processes (and if there is an SE organization, the organization) within and outside the systems community, including to project managers, other engineers, and clients.

**13 ED Educator.** A systems engineering educator brings knowledge in systems engineering techniques, practices, processes, or background knowledge to others. This role could be fulfilled by a formal academic or adjunct professor who teaches classes to undergraduate or graduate students; organizational staff who bring together systems engineers to share lessons in informal settings such as lunch-and-learn meetings; or even an AI/ML module or chatbot who helps young staff learn systems engineering practices, techniques, or processes in real time.

Many systems engineers frequently serve as educators. Some believe the only good application of systems engineering is to make systems engineering become "the way engineering is done" across the organization (Beasley, 2022); these people make it their job to perform the ED role continually.

**14 IE Implementation Engineer.** Systems engineers do not build elements, but one could argue they do build systems. To the extent they do, it is the Implementation Engineer role that does the building. Software engineers implement software systems, which generally means they write code, or else find close-to-need modules in code repositories, then modify them to make the system work. The need includes security and timing constraints as well as significant other architectural constraints that make finding an appropriate module far from trivial. Frequently, even if an appropriate core module is found to do the major calculation, enough wrappers and interface software must be developed that the job is not much easier than building new software.

There is a significant chance that in the not-too-distant future an actual system implementation role will be defined for systems engineers. In particular, it is becoming increasingly clear that many of the steps currently performed by engineers will be able to be done by artificial intelligence. Thus, implementation of the system itself may be possible by systems engineer(s), when formerly implementation had to be done by a cadre of specialist engineers.

**15 IM Information Manager.** The information manager role is the systems engineering part of information management. The amount of information being generated, used, and managed has exploded since 1996. This role ensures the appropriate system-relevant information (including technical, cost, and schedule information) is identified, collected, and distributed to the appropriate people in time for decision-making. Whoever plays this role needs to be versed in data governance and model governance, to reduce the chances of contagion and propagation of poor-quality information or errors across the digital engineering ecosystem. The Information Manager role also supports system integrity by assuring there are actively followed processes that ensure information correctness and timeliness.

This role may expand into the data hygiene and management side of AI in the next two years, if it isn't there already. (Sheard, 2025).

The Information Manager must interface with whichever roles deal with data, particularly RO for requirements information, MO for all their data and models, SA and SD regarding system data that is outside of any models, TM regarding project status, and VV regarding test data.

**16 IN Innovator/Initiator.** Organizations sometimes find they have engineers who frequently offer innovative ideas for products and/or for processes, and initiate activities that challenge the status quo toward better activities and outputs. Dynamic and healthy organizations make use of these internal innovators, turning their ideas into new products, processes, and patents. Such organizations can make technical advances that leapfrog competitors and have the agility to withstand changes in business environments.

The more entrepreneurial of the Innovators can first make a business case for the idea by identifying costs, pricing, and rates, and after approval can establish financial reporting and metrics, then carry through with project execution given ongoing internal support from company leadership. Working with the leader role, the entrepreneur role can transition into a manager role.

Hutchison et al.'s (2017) role of "concept creator" includes such actions where they would lead to a program contract. IN is also similar to SA of an early program.

**17 LC Life Cycle Engineer.** This role is responsible for ensuring the life cycle perspective is honored in all phases of the system life cycle. This includes ensuring sustainability, transition, maintenance, disposal, and even public perception during operation are considered. If systems engineers are involved in the operational phase of the system life cycle, perhaps operating the system for the customer, or serving on call to answer questions and resolve anomalies, they do so in this role.

To the extent that the system includes the digital engineering environment (DEE) and software widgets, the Life Cycle Engineer is always working with their operational aspects. In any case the LC role deals with the DEE operation.

**18 PO Process Owner.** As process owners, systems engineers are expected to document, own, evaluate, and improve the projects' and the organization's systems engineering-related processes. The Process Owner helps projects select organizational processes and tailor them for project needs. Among the complexities of system and software processes are integrating agile lifecycle-based processes for software development with any incremental processes needed for hardware, and implementation of processes such as SAFe (Scaled Agile Framework) (Leffingwell, 2016) and DevSecOps (development-&-operations-with security) (Turner, 2021).

In addition, processes must be flexible to accommodate today's constant upgrades and changes. Today a Process Owner can implement validation rules that automatically check digital artifacts to ensure established processes are being followed. In the future, this may include increasingly adopting and improving AI that checks compliance to processes or even automatically completing the more automatable parts of the processes.

## Derivation of the 18 Roles from Literature

The new roles above are consolidated primarily from the original paper and from four additional sources discovered from the literature search. Chief among the sources is (Græssler et al., 2019), which did a thorough job of updating the roles into usefulness with the 21st century and socio-technical cyberphysical systems. Græssler also integrated five competency models and the 2015 version of the INCOSE Systems Engineering handbook. Hutchison, et al. (2017) made significant progress in getting community feedback on the meanings of the role names and their usefulness in the real world.

Table 4 below shows the relationship of these 18 roles to (Sheard, 1996), (Sheard, 2000), (Amit, et al., 2013), (Hutchison, et al., 2017), and (Graessler, et al., 2019). Role 11, AI Chief, is derived from a 2024 source as mentioned.

The biggest changes were the long-required addition of a Security Engineer (abbreviation CY for cybersecurity), a System Modeler (MO) to upgrade system analyst to a digital and model-based world, and an AI Chief to acknowledge the need to have someone lead the interface between a project and everything that is changing on a daily basis in the AI world.

The original lifecycle roles (RO, SD, VV, LO, and SA=system analyst) are still there, as Primary roles except for LO. System Designer has separated into Systems Architect and System Designer, System Analyst has morphed into System Modeler, and RO and VV have become more modern RO and VV. Logistics and Operations (LO), now Life Cycle Engineer, has become a secondary role. As originally stated, LO comprised whatever systems engineers did in the later phases of programs, plus bringing the later-stage perspectives to requirements, design, and development phases. Life Cycle Engineering perspectives are something that all the primary systems engineers need to have, so it is fair to make the rest a support role.

Readers of the original paper tended to either love or hate the term *Glue,* as it was evocative rather than descriptive. The descriptive terms *interface* and *integration* were not used back then because they were overloaded, bringing to mind specific contractors rather than activities. Now, parallel and separate terms *System* and *Stakeholder* Interface Owners allow a return to descriptive terms to replace G (Glue) and CI (Customer Interface). The concept of integration is now integral to SI (System Interface Owner).

Other original roles have evolved but not in a major way. Minor additions include Implementation Engineer, with exact amount of effect and role definition to be determined, Champion (CH), Educator (ED), and Innovator/Initiator (IN), to be applied as appropriate.

## Applying Systems Engineering Roles

### *Basic Roles*

In any system development project, basic systems engineering roles need to be fulfilled. The primary systems engineering roles were created to be those basic roles. All systems will need requirements, analysis via modeling, architecture and design, attention to interfaces both external and internal, proof via validation

Table 4. Connection of New Roles to Sources

| | **↓2025 Role Source →** | * | **(Sheard, 1996)** | **(Sheard, 2000)** | **(Amit, 2013)** | **(Hutchison 2017)** | **(Græssler, 2019)** |
|---|---|---|---|---|---|---|---|
| ***Primary Roles*** | | | | | | | |
| **1 CL** | **Coordinator/Leader** | T | CO | | | CO | L, IE |
| **2 CY** | **Security Engineer** | S | | | | | Secur |
| **3 MO** | **System Modeler** | S | SA†, SD VV | | | SysAna | Model, StIM SIM, TM |
| **4 RO** | **Requirements Owner** | S | RO | | | RO | RE |
| **5 SA** | **Systems Architect** | S | SD | | Arch | SysArch | SysArch |
| **6 SD** | **System Designer** | S | SD | | | DD | SysArch |
| **7 SI** | **System Interface Owner** | S | G, SD | | | SI | SIM |
| **8 ST** | **Stakeholder Interface Manager** | T | CI | | | CI | StIM |
| **9 TM** | **Technical Manager** | T | TM | | | TM | TM |
| **10 VV** | **Validation and Verification Engineer** | S | VV | | | VV | VV |
| ***Secondary Roles*** | | | | | | | |
| 11 AI | AI Chief** | O | | | | | |
| 12 CH | Systems Engineering Champion | T | | SEE | | SEC | |
| 13 ED | Educator | T | | ED | | Inst./Tch | |
| 14 IE | Implementation Engineer | S | CA | | | | |
| 15 IM | Information Manager | S | IM | | | IM | IM, CM |
| 16 IN | Innovator/Initiator | O | | | Inn/Ini | CC | Entrepreneur |
| 17 LC | Life Cycle Engineer | S | LO | | | Support | LCE |
| 18 PO | Process Owner | O | PE | | | PE | PO |

*Hutchison et al.'s focuses of System, Organization/Process or Team. **(Cummings, 2024)
†Note, the 1996 paper used "SA" for System Analyst, not System Architect.

and verification, and some sort of coordination and management. Any system today will need security and privacy awareness. The extent to which these roles will be done, and by whom, will have to be defined.

The secondary roles round out the roles considered by some to belong to systems engineering. Not all implementations will have all, or maybe any, of these. They are included so that negotiations as to whether they are to be done, included in expectations, or performed in a contract, can use the same vocabulary.

## Roles for a Job or Position

**Creating a position in an organization.** When defining roles within an organization or project, functional roles and responsibilities will be set up that are specific to that instantiation. For example, "Jackie Oxmore will be Chief Architect on the Solar Sail for the X project." While one can assume Jackie will be responsible for most or all of the Systems Architect role above, there will be specific additional duties stated for Jackie's job. Jackie will have reporting duties, and likely have certain financial responsibilities, due dates, and other job responsibilities that are not specifically systems engineering.

Every job or position has these characteristics. An organization has needs which it hires people to fulfill. No job is an exact copy of a systems engineering role in this paper. Rather, it may be an application or tailoring of one or more of these roles to a specific organization, and will come with specific organizational boundaries, schedules, budgets, and reporting responsibilities.

The basic systems engineering roles are things people generally need to do to make a good system. It is up to the organization to determine how to split the roles among the people they have to get the job done. The organization must create positions that will meet the organization's needs, sometimes called functional roles and responsibilities.

**Example.** As Cummings (2024) noted a critical need for an AI Chief, others have asked for a similar roles within systems engineering at the enterprise level. A comparison of the basic roles to the UK Government Digital and Data Profession Capability Framework (D. Beale, personal communication) suggests that there is a need for an Organizational Systems Engineer role, comparable to an Enterprise or Business Architect role, that applies the basic systems engineering roles but with more of an enterprise architect or business focus. This role also needs to focus more on the people element, considering culture and behaviors; these are typical considerations for sociotechnical systems. (Baxter & Sommerville, 2011) As a critical need, this is up to the organization to establish and fill. Note that INCOSE has done some work on the interaction between business and systems engineering; see, for example, (Siviy, 2023). The UK Organizational Systems Engineer role should be a Systems Architect role, with focus on people and business aspects, possibly combined with parts of other roles including Process Owner and Information Manager roles.

## Systems, Teams, and Organizations

Hutchison, et al. (2017) assigned each role to whether it primarily focused on the system being developed, the teams building the systems, or the process and organizations. Those are indicated by the S, T, and O in the third column of Table 4 (the O for AI, and the S for IE, were added by this paper's authors.)

## Some Notes

**Configuration Management.** Configuration Management (CM) is considered a role by some, and in government it is a functional role assigned within the systems engineering contract. Here it is considered part of the Information Manager role, although today's data needs are increasing quickly, and some therefore suggest a separate Data Manager role. As AI booms, the IM role expands considerably and CM cannot be neglected, so some (Graessler, 2019) have called out CM separately from IM. The information produced and managed in the various roles has similar considerations, though.

**Agile**. Agile project management is related to many SE roles, including CY, RO, MO, TM, and VV, with IE involved as well. SA and SD may be somewhat undercover in Agile development, as design may take a back seat to making progress today. It is important, though, to ensure that nothing done in real time during agile software development "breaks" the system architecture or design (or if it does, that a technical debt is recorded to fix it later). The authors consider agile considerations to be one method of implementing SE, in which case, it may fall within the scope of the Life Cycle Engineer role.

## Conclusions

Clearly defined roles are important because they form a base on which various people and organizations can establish who is going to be responsible for what in creating and sustaining a system, whether a new system, a newly recognized system, or an ongoing system. For nearly thirty years an eight-page paper presented in an INCOSE conference has stimulated thousands of readers to discuss "What is systems engineering?" Ideally the purposes to which the original roles have been put (see Table 1) can be better served now that the roles have been updated to consider the way systems engineering is currently done, including MBSE, digital engineering, and AI.

Table 2 shows the 2025 systems engineering roles. The twelve roles from 1996 all still exist but have been updated to accommodate industry changes. Ten roles are primary – required for all systems engineering efforts – and eight are support. System Modeler and Security Engineer are major roles that have been added. The 1996 System Designer has been separated into Systems Architect and System Designer. Some roles have remained mostly the same but have changed names, mostly for clarity. AI Chief, Systems Engineering Champion and Educator have also been added as support roles. Hutchison et al. (2017) grouped these into those that focus on the systems being developed, the organization or process, or the team building the system; Table 2 shows which roles fit in which categories.

The 2025 roles have many uses in terms of determining processes, career paths, organizational capabilities, capability models, research agendas, or training programs.

## Recommendations

The new systems engineering roles should be used instead of the original twelve, as these have been adapted to the changed environment of systems engineering since 1996. The potential future uses of SE roles include continuing to use the roles as in Table 1 plus additional uses shown in Table 5. Uses of the roles as the industry switches over to intense use of AI still leave a lot to be determined, but human intelligence supported by AI will grow, and determining the roles each can play is a way to see through the confusion.

Table 5. Roles Can Helped Solve Future Problems

| Problems In the Future | How Roles Can Help Solve Them |
|---|---|
| Individual organizations need help assigning new people transitioning in and out | Define work by career paths vs new systems engineering roles and competencies, so new people can quickly place where they are and where they want to be (Græssler et al, 2024) |

| | |
|---|---|
| Lower availability of skilled engineers in the workforce | Pair AI roles with humans to offset shortage. (Subrahmanyam, 2023) |
| A particular large collaboration needs to have stable workforce despite changing times and people | Agree who does what roles for this collaboration using the updated roles. |
| Standards and other models (e.g., Digital engineering competency model, Scaled Agile Framework) change to implement more digital engineering and more AI | Organize the specifications by roles that will not change. These roles, while updated after 30 years, are actually fairly stable. |
| Large collaborations need to involve more disciplines to make United Nations challenges happen | Build a picture of the roles within the larger project and how the disciplines performing their role is critical to meet the challenges. Do not make this all about systems engineering, but rather make it about the challenge. Use SE knowledge to downplay the role of SE. |
| Organizations need people with the right skills performing the right roles | Use role descriptions that map to competency models. (Græssler, 2024) But don't be too optimal; systems engineers need to have broad experience too. |
| Organizations need to partner with AI across a large project | Per (Sheard, 2025), SE should supervise AI when ethical decision making is required, in high-stakes environments, and for management of complex systems. SE and AI should partner in healthcare, creative fields, and in customer service. AI should take charge in data analysis, routine monitoring, and predictive maintenance. Set up AI for what it does best and set humans in the right roles for them. |
| Need new research agendas | Plan some research about how best to accomplish each of the new SE roles, per (Erasmus, 2013) |


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## Biography

**Sarah Sheard** spent her 30+ year career moving between process improvement, systems engineering, and software engineering (now venturing into the impact on these of AI); between practicing, managing, consulting, writing about, and teaching; and with industry (satellites), government (air traffic control) and academia. She joined INCOSE in 1992, became a fellow in 2006, and won the Founder's award in 2002. Her papers won "best paper" awards five times, first in 1992 and most recently in 2021. She received her PhD on Complex Systems in 2021 and since retiring in 2019 she has continued to serve in INCOSE working groups.

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